SonarIQ Demo ship · Rev A Live

Profile: Passenger ship · > 36 passengers

Scope: Selection | **Deck 5** | Whole model

Packs: ☑ Fire & life safety ☑ Accessibility ☑ Class · DNV To verify

Analyse

– Fail	– Warning	– Cannot verify	– Pass	– Dormant

Findings | Chat | History

96 checks in 3 packs will run on Deck 5.

The rule engine decides every verdict. It needs no AI and no network.

Fire & life safety · SOLAS II-2, FSS Code Ch.13	64	Verified
Accessibility · PVAG 2013 draft, ADA 2010	20	Verified
Class · DNV	12	To verify

Class rules marked To verify still run. Their verdicts carry the To verify tag until the pack is signed off.

No findings yet

Press Analyse. Findings highlight the objects in the 3D view first, then list here.

`Ctrl Shift A` Analyse

`Live` re-checks what each change touches, as you work

`Scope` select objects in the tree to check only those

IFC4 · Demo_ship_RevA Rule engine ready

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Analysing… 61 of 96 Cancel

1	0	1	57	2
Fail	Warning	Cannot verify	Pass	Dormant

Findings 2 Chat History

Analysing Deck 5 61 of 96 · 1.3 s

Fire & life safety 49 / 64 · running

Accessibility 12 / 20 · running

Class · DNV (To verify) 0 / 12 · queued

FOUND SO FAR 2

Fail F-01 **Fire door class below bulkhead**

Deck 5 · Fr 165 · P

SOLAS | II-2/9.2.2.1.1

Cannot verify A-01 **Door has no fire class in the model** new

D-5-171-S Deck 5 · Fr 171 · S

Findings appear as each check finishes. Counts settle when every pack is done.

Fire & life safety and Accessibility running · Class queued

Cancel keeps the 61 results so far, marked as a partial run.

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Packs: ☑ Fire & life safety ☑ Accessibility ☑ Class · DNV To verify

Analyse 96 checks · 2.1 s · Rev A · 14:20

1	1	1	89	4
Fail	Warning	Cannot verify	Pass	Dormant

Findings 3 Chat History

Showing open findings 3 Group: **Discipline** | Location

FS Fire & life safety 2

Fail F-01 **Fire door FD-5-165P below bulkhead class**

Deck 5 · Fr 165 · P

SOLAS II-2/9.2.2.1.1

MVZ 4/5 bulkhead · fitted A-0 , required A-60

Cannot verify F-01 **Door D-5-171-S has no fire class**

Deck 5 · Fr 171 · S

add the property or confirm

AC Accessibility 1

Warning W-01 **Corridor below width guideline**

Deck 5 · Fr 152 · S

PVAG 2013 draft

clear width 900

mm

, guideline

915 mm (36 in)

· draft, so a warning

CL Class · DNV To verify 0

No open findings in this pack.

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Packs: ☑ Fire & life safety ☑ Accessibility ☑ Class · DNV To verify

Analyse 14 re-run · 0.4 s · Rev B · 14:32

4	1	1	86	4
Fail	Warning	Cannot verify	Pass	Dormant

Findings 6 | Chat | History

← Findings F-07 · 2 of 6 open ↑ ↓

Fail F-07 Rule engine

Stair 3B is narrower than the people it serves need

Stair 3B · Decks 4–7 Deck 5

FS Fire & life safety

FSS | Code Ch.13 §2.1.2 sample rule text

> Stairway clear width shall not be less than W, worked out from the persons served on each deck the stair connects.

REQUIRED	MEASURED	MARGIN
1,380 mm	1,350 mm	−30 mm

FORMULA · INPUTS FROM THE MODEL

$$W = (N_1 + N_2 + 0.5\,N_3 + 0.25\,N_4) \times 10 \text{ mm}$$

$$= (\ 68_{N_1\ D5} + 44_{N_2\ D6} + 0.5 \cdot 40_{N_3\ D4} + 0.25 \cdot 24_{N_4\ D7}\ ) \times 10$$

$$= (68 + 44 + 20 + 6) \times 10 = 138 \times 10 =$$

1,380 mm

Highlighting in 3D: Deck 5 · 68 persons: 60 before OR-117, + 8 berths in the four new cabins.

SonarIQ Demo ship · Rev B Live

Profile: Passenger ship · > 36 passengers

Scope: Selection | **Deck 5** | Whole model

Packs: ☑ Fire & life safety ☑ Accessibility ☑ Class · DNV To verify

Analyse 14 re-run · 0.4 s · Rev B · 14:36

3	1	1	87	4
Fail	Warning	Cannot verify	Pass	Dormant

Findings 5 | Chat | History

Rev A · Analyse · 14:20

YOU

Add four cabins aft of frame 150

EDIT PREVIEW AI draft · you decide

Action	Add cabin block × 4
Where	Deck 5 · aft of Fr 150
Berths	8
Change order	OR-117

Confirm Cancel Nothing changes until you confirm.

You confirmed OR-117 · Rev B · 14:32

14 checks re-run · 3 changed 0.4 s

Fail F-07 Stair 3B · Decks 4–7 was Pass

Fail F-08 Dead-end · Fr 146–150 · P was Pass

YOU

Why does 3B fail, and what can I do?

✓ AI explanation · numbers checked

860 px panel fold · content unrolled below

Stair 3B needs 1,380 mm of clear width and has 1,350 mm, so it is 30 mm short.

Its width comes from the people each deck sends to it: $W = (N_1 + N_2 + 0.5\,N_3 + 0.25\,N_4) \times 10$ mm.

Your change added 8 berths on Deck 5, so N_1 went from 60 to 68. Decks 6, 4 and 7 are unchanged (N_2 44, N_3 40, N_4 24). In Rev A the stair needed 1,300 mm.

The stair serves Decks 4–7, so the failure sits on decks nobody touched.

To pass, the stair gets wider or Deck 5 sends fewer people to it. Each option below was re-checked by the rule engine.

1 · Widen Stair 3B to 1,400 mm Pass

1,380 ≤ 1,400 mm · margin 20 mm

ST Needs Structure review deck openings, Decks 4–7

Apply Preview in 3D Re-checked

2 · Keep four cabins, one berth each Applied

N_1 64 → 1,340 ≤ 1,350 mm · margin 10 mm

No structural change. Sets 1 berth on each of the 4 cabins in OR-117.

Applied by you · 14:36 · Rev B · Undo until the next save

3 · One cabin instead of four Pass

N_1 62 → 1,320 ≤ 1,350 mm · margin 30 mm

Changes the scope of OR-117.

Apply Preview in 3D Re-checked

▸ 1 idea rejected by the engine

Fix 2 applied · Rev B · 14:36

Pass F-07 **Stair 3B** Rule engine

1,340 ≤ 1,350 mm · margin 10 mm

Still open from this change: F-08 dead-end corridor, F-09 corridor pinch.

Accept F-08 with reason... Open F-09

What did my last change affect? Why does stair 3B fail?

Ask a question or describe a change Send

Edits I can make: add a cabin block · move a wall · set a door's width · set a door's fire class.

SonarIQ Demo ship · Rev B Live

Profile Passenger ship · > 36 passengers

Accept F-08 with reason

Fail Dead-end corridor Rule engine

Deck 5 · Fr 146–150 · P SOLAS II-2/13.3.1.2

Depth 4.2 m, width 1.2 m. A recess is allowed only if depth ≤ width.

Reason required

[object Object]

Example: what the space will become and why the risk is controlled.

Rule	SOLAS II-2/13.3.1.2 · stays Fail in the engine
Goes to	FS Fire & life safety lead Mikko Saari
Signed as	Aino Laine · Outfitting designer · Rev B

WHAT HAPPENS NEXT

1. F-08 shows Accepted with reason · awaiting approval.
2. The lead approves, requests a change, or declines with a reason.
3. Declined returns it to you as Fail. Any edit to this corridor re-opens it.

Send for approval Cancel

Profile Passenger ship · > 36 passengers

Send Rev B to review

DECISIONS · 2

- Pass F-07 fix 2 applied 14:36
- Fail F-08 accepted with reason · needs approval 14:41

OPEN FINDINGS · 4

- Fail F-01 FD-5-165P · A-0 in A-60
- Fail F-09 corridor pinch · 850 mm
- Warning W-01 850 mm vs 915 mm
- Cannot verify U-01 D-5-171-S

RECIPIENTS BY DISCIPLINE

- FS Fire & life safety 5 items
- AC Accessibility 1 item

Structure, Architecture, HVAC, Systems: nothing from this change.

sonariq_rev-b.bcfzip
BCF 3.0 · 6 topics · opens in any BCF-aware design tool

Change order OR-117 is linked. Each item is signed with your name, role, time and Rev B.

Send to review Cancel

SonarIQ Demo ship · Rev B Live

Profile: Passenger ship · > 36 passengers

Scope: Selection | **Deck 5** | Whole model

Packs: ☑ Fire & life safety ☑ Accessibility ☑ Class · DNV To verify

Analyse 14 re-run · 0.4 s · Rev B · 14:36

3	1	1	87	4
Fail	Warning	Cannot verify	Pass	Dormant

Findings 5 | Chat | History

Rev B · 14:44

YOU

Raise the deckhead on Deck 5 by 100 mm

I can't make that edit Edit guard

Deckhead height is not one of the edits SonarIQ is allowed to make. Your model is unchanged.

EDITS I CAN MAKE

Add a cabin block	"add 2 cabins aft of Fr 150"
Move a wall	"move the wall at Fr 152 · S 50 mm"
Set a door's width	"set D-5-171-S to 900 mm"
Set a door's fire class	"set FD-5-165P to A-60"

WHAT I CAN DO INSTEAD

Make the change yourself in FreeCAD. With Live on, the rule engine re-checks what it touches and tells you what changed.

Turn Live on | Which rules use deckhead height?

The guard is not AI: the request is parsed, matched against the four

What did my last change affect? | Why does stair 3B fail?

Ask a question or describe a change Send

SonarIQ Demo ship · Rev A Live

Profile: Passenger ship · > 36 passengers

Scope: Selection | **Deck 5** | Whole model

Packs: Fire & life safety | Accessibility | Class · DNV — To verify

Analyse — 96 checks · 2.1 s · Rev A · 14:20

1 Fail	1 Warning	1 Cannot verify	89 Pass	4 Dormant

Findings 3 | Chat | History

← Findings U-01 · filter: Cannot verify

Cannot verify U-01 Rule engine

Door D-5-171-S has no fire class in the model

Deck 5 · Fr 171 · S D-5-171-S

FS Fire & life safety

WHAT IS MISSING

Property	Pset_DoorCommon.FireRating
Value in model	empty
Needed by	SOLAS — II-2/9 · door fire integrity
Waiting on it	1 check

SonarIQ does not guess. A missing class is not treated as A-0 or as a pass; the check stops here until someone supplies the value.

RESOLVE IT · PICK ONE

1 **Add the property in FreeCAD**

Select the door › Property view › IFC › FireRating. Live re-checks it on save.

Select D-5-171-S in tree

2 **Set the fire class here**

An allowed edit. You see a preview and confirm before

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Analyse 96 checks · 2.1 s · Rev A · 14:20

1	1	1	89	4
Fail	Warning	Cannot verify	Pass	Dormant

Findings 3 Chat offline History

Chat is offline

The AI endpoint is not reachable. Explanations, typed edits and fix proposals are paused.

On-premises endpoint · no reply since 14:52 · retrying every 30 s

Retry now AI endpoint settings

STILL WORKING · THE RULE ENGINE NEEDS NO AI

- ✓ Analyse and Live re-checks
- ✓ Findings, 3D highlights and pins
- ✓ Evidence: rule, formula, required vs measured
- ✓ Accept with reason, Send to review, BCF export
- ⊟ Paused: explanations, typed edits, fix proposals

Your earlier conversation is kept and returns when the endpoint is back. Nothing is queued to send while offline.

Last AI reply · 14:51

Chat is offline. Analyse still works. Send

Go to Findings

A · COMPACT WHILE YOU WORK

SonarIQ Rev A — Live — Expand

- Live · last check 2 s ago · 0 new

1 Fail · 1 Warning · 1 Cannot verify · 89 Pass · 4 Dormant

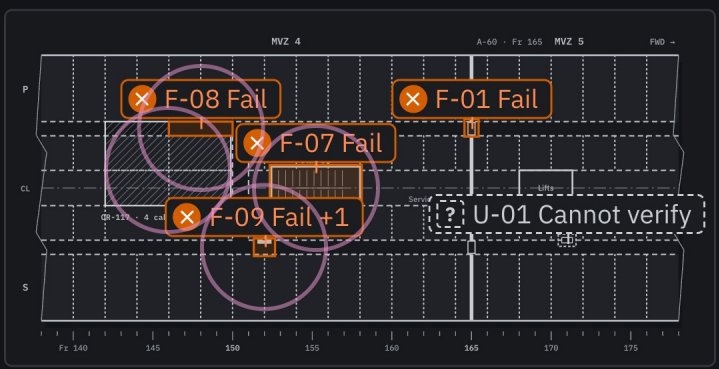
B · THE PING AS IT LANDS

SonarIQ Rev B — Live — Expand

14 checks re-run · 3 changed — just now

4 Fail (+3) · 1 Warning · 1 Cannot verify · 86 Pass (−3) · 4 Dormant

Fail	F-07 Stair 3B · Decks 4–7	was Pass
Fail	F-08 Dead-end · Fr 146–150 · P	was Pass
Fail	F-09 Pinch · Fr 152 · S	was Pass

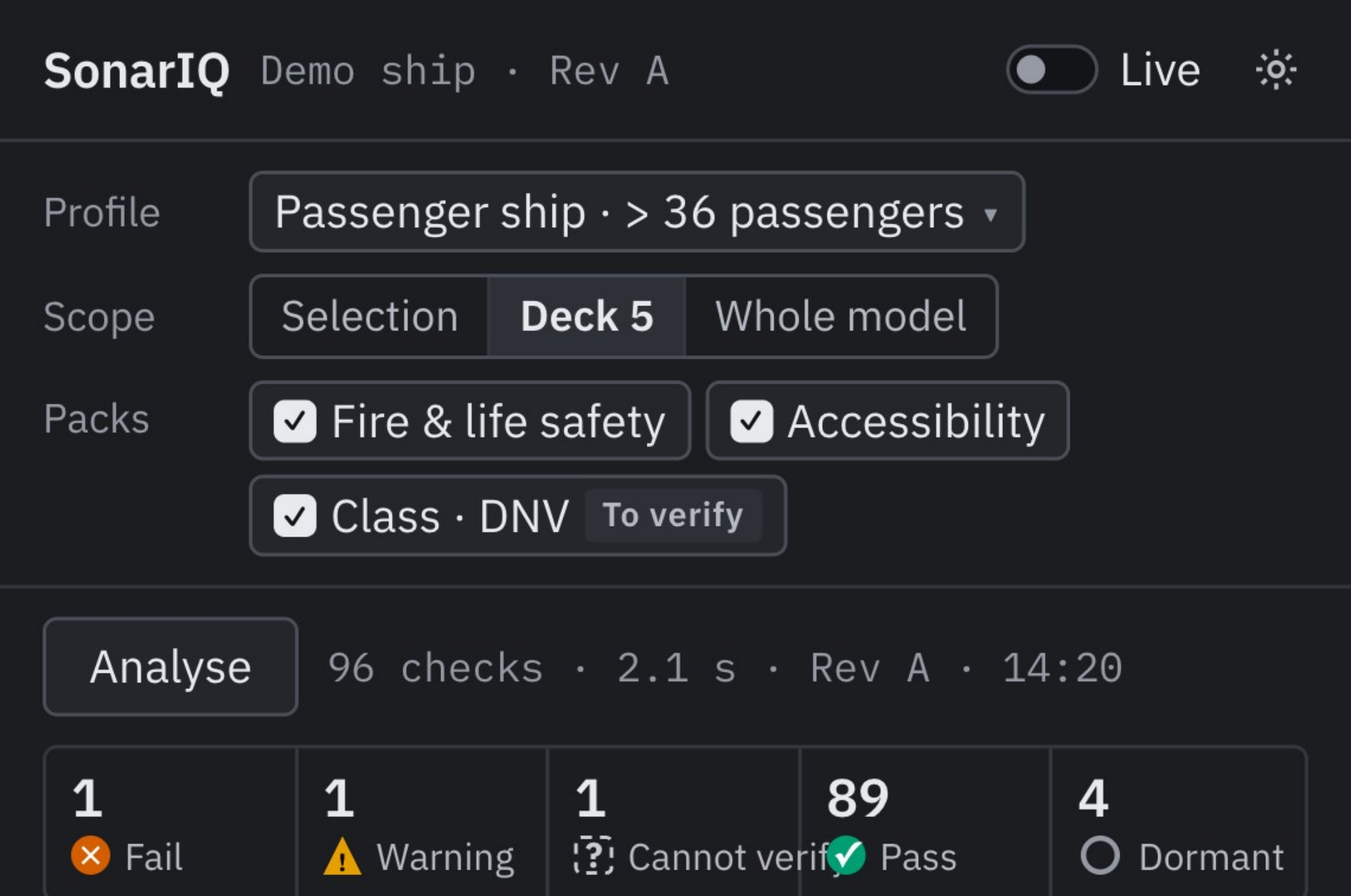


3D: one ring from each element whose verdict changed, plus the new block. 600 ms, ease-out, once. Counts update as the ring starts; changed counts keep the accent tint for 2 s.

C · REDUCED MOTION

14 checks re-run · 3 changed

No ring in 3D. Fail and Pass counts hold the accent tint; the three changed rows are listed. Screen readers hear: “3 checks changed to Fail.”


SonarIQ
Demo ship · Rev A
Live
Profile
Passenger ship · > 36 passengers
Scope
Selection
Deck 5
Whole model
Packs
Fire & life safety
Accessibility
Class · DNV
To verify
Analyse
96 checks · 2.1 s · Rev A · 14:20
1
Fail
1
Warning
1
Cannot verify
89
Pass
4
Dormant

Fail F-07 Rule engine

Stair 3B is narrower than the people it serves need

Stair 3B · Decks 4–7 | Deck 5

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$$= (68 + 44 + 20 + 6) \times 10 = 138 \times 10 = \mathbf{1{,}380\ mm}$$

Highlighting in 3D: Deck 5 · 68 persons: 60 before OR-117, + 8 berths in the four new cabins.

Rev A needed 1,300 mm (N_1 was 60). The stair spans four decks nobody touched.

HOW IT WAS MEASURED

Clear width between handrails, from IFC geometry.